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CHE 301 Final Formulas

Physical Chemistry I (Stony Brook University)



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CHE 301, Fall 2022
Practice Final Exam

This exam is closed book and closed notes. You may use a calculator. You may not use a cell phone or communicate with others during the exam. **WRITE ALL YOUR ANSWERS IN THE BLUE EXAM BOOK**, and write your **name, exam version or color, and ID number** on the front of the blue book. Please also write your name and ID on the exam, and turn in your copy of it.

There are **NINE** problems in the exam. **BOTH THE EQUATION SHEET AND THE EXAM SHEETS ARE DOUBLE-SIDED**. Show your work and, if you are running short of time, try to explain what you would do to solve the problem. Please circle or box your final answers to all problems. Additional information that may be useful is on the first page of the exam. You may tear off this top page with equations while you work on the exam.

Energy, enthalpy, heat, and work

$\Delta U = q + w$ (closed system)

$H = U + PV$

$q = \int_{T_1}^{T_2} C\, dT$

$w = -\int_{V_1}^{V_2} P_{ex}\, dV$ (Expansion of a gas)

Gases

(assuming ideal gas, C_p and C_v independent of T)

$\Delta U = nC_{v,m}(T_2 - T_1)$

$\Delta H = nC_{p,m}(T_2 - T_1)$

$C_{p,m} = C_{v,m} + R$

Isobaric process:

$q_p = nC_{p,m}(T_2 - T_1)$

$w_p = -P(V_2 - V_1) = -nR(T_2 - T_1)$

$\Delta S = nC_{p,m} \ln \frac{T_2}{T_1}$

Isochoric process:

$q_v = nC_{v,m}(T_2 - T_1)$

$w_v = 0$

Isothermal process:

$q_T = -w_T = nRT \ln \frac{V_2}{V_1} = nRT \ln \frac{P_1}{P_2}$

$\Delta S = -nR \ln \frac{P_2}{P_1} = nR \ln \frac{V_2}{V_1}$

Adiabatic process:

$w = nC_{v,m}(T_2 - T_1)$

$q = 0$

$C_{v,m} \ln \frac{T_2}{T_1} = -R \ln \frac{V_2}{V_1}$

Chemical Potential and Activity

$\mu_A = \mu_A^o + RT \ln a_A$

Ideal gas: $a_A = P_A$ (P_A in atm or bar)

Real gas: $a_A = \gamma_A P_A$ (P_A in atm or bar)

Pure liquid: $a_A = 1$

Pure solid: $a_A = 1$

Solvent: $a_A = \gamma_A x_A$ (x_A is mole fraction)

Solute: $a_B = \gamma_B c_B$ (c_B in molar units, M)

Ideal solution, solute: $a_B = c_B$ = [B]

Ideal solution, solvent: $a_B = x_B \approx 1$

Chemical Reactions and Chemical Equilibrium

For aA + bB → cC + dD at constant T and P:

$\Delta G^o{}_{rxn} = c\,\overline{G_C^o} + d\,\overline{G_D^o} - a\,\overline{G_A^o} - b\,\overline{G_B^o}$

$\overline{G^o}_i = \Delta G^o{}_{f,i}$

$\Delta H_{rxn} = c\overline{H_C} + d\overline{H_D} - a\overline{H_A} - b\overline{H_B}$

$\overline{H^o}_i = \Delta H^o{}_{f,i}$

If ΔH_{rxn} is constant over the temperature range:

$\frac{\Delta G_{rxn}(T_2)}{T_2} - \frac{\Delta G_{rxn}(T_1)}{T_1} = \Delta H_{rxn}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$

$\Delta G^o{}_{rxn} = -RT \ln(K)$

$\Delta G_{rxn} = \Delta G^o{}_{rxn} + RT \ln(Q)$

$K = \frac{\left(a_C^{eq}\right)^c \left(a_D^{eq}\right)^d}{\left(a_A^{eq}\right)^a \left(a_B^{eq}\right)^b}$

$Q = \frac{(a_C)^c (a_D)^d}{(a_A)^a (a_B)^b}$

MORE EQUATIONS ON THE BACK →

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Temperature Dependence of *K*

If ΔH^{o}_{rxn} is constant:

$$\ln\frac{K_2}{K_1} = -\frac{\Delta H^{o}_{rxn}}{R}\left(\frac{1}{T_2}-\frac{1}{T_1}\right)$$

Phase Equilibria and Osmotic Pressure

Raoult's Law

$P_A = x_A P_A^{vap}$ or $y_A P = x_A P_A^{vap}$

Henry's Law

$P_B = x_B\, k_B$ or $y_B P = x_B\, k_B$

Transfer from Phase 1 to 2, uncharged species

$$\Delta G = -nRT\ \ln\frac{a_{A,phase\,2}}{a_{A,phase\,1}} \cong -nRT\ \ln\frac{c_{A,phase\,2}}{c_{A,phase\,1}}$$

For neutral, non-dissociating species

$\Pi = cRT = \frac{wRT}{M}$, where

w = concentration in kg/m^3

M = molecular weight in kg/mol

Π = osmotic pressure in Pa

Molecular Motion

$$F(v)dv = 4\pi\left(\frac{m}{2\pi kT}\right)^{3/2} v^2 e^{-mv^2/2kT}dv$$

$$v_{ave} = \sqrt{\frac{8kT}{\pi m}} = \sqrt{\frac{8RT}{\pi M}}$$

$$v_{rms} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3RT}{M}}$$

$$v_{mp} = \sqrt{\frac{2kT}{m}} = \sqrt{\frac{2RT}{M}}$$

First-Order Kinetics

$$-\frac{d[\mathrm{A}]}{dt} = k_1[\mathrm{A}]$$

$$\ln\frac{[\mathrm{A}]}{[\mathrm{A}]_o} = -k_1 t$$

$[\mathrm{A}] = [\mathrm{A}]_o \exp(-k_1 t)$

$$\ln\frac{[\mathrm{A}]_2}{[\mathrm{A}]_1} = -k_1(t_2 - t_1)$$

$$t_{1/2} = \frac{\ln 2}{k_1}$$

Temperature Dependence of Rate Constants

Arrhenius expression:

$k_1 = A \exp(-E_a/RT)$

$$\ln\frac{k_2}{k_1} = -\frac{E_a}{R}\left(\frac{1}{T_2}-\frac{1}{T_1}\right)$$

Enzyme Kinetics

Michaelis-Menten Mechanism:

$$v_0 = \frac{V_{max}}{1+\frac{K_M}{[S]}}$$

$V_{max} = k_2[\mathrm{E}]_0 \quad K_M = \frac{k_{-1}+k_2}{k_1}$

Quantum Mechanics

(for clarity, f is frequency and v is velocity)

$\lambda f = c$

$E = hf$

$$\lambda = \frac{h}{mv}$$

$$\Delta p \Delta x \geq \frac{1}{2}\left(\frac{h}{2\pi}\right)$$

$$\delta E \approx \frac{h}{2\pi\tau}$$

Physical Constants

R = 8.31451 J/mol K = 0.08205 L atm/mol K

N_o = 6.022 x 10^{23} 1/mol

k = 1.3807 x 10^{-23} J/K

c = 2.9979 X 10^8 m/s

h = 6.626 x 10^{-34} J s

Conversion Factors

K = ^{o}C + 273.15

1 kg = 1000 g

1 g = 1000 mg

1 m = 10^{10} Å = 10^9 nm = 100 cm

1 m = 39.37 in = 3.2808 ft

1 L = 1000 mL = 1000 cm^3

1 Pa = 1 N/m^2 = 14.696 lb$_f$/in^2

1 atm = 760 torr = 1.013 x 10^5 Pa

1 J = 1 N m = 1 Pa m^3 = 0.23901 cal

1 eV = 1.602 x 10^{-19} J

1 Hz = 1 s^{-1} = 1 x 10^{-6} MHz

1 kJ = 1000 J

1 M = 1000 mM = 1 x 10^6 μM

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